pr1-avanzada

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2026-02-11

1 dab

1.1 dab

1.2 Qualitat de les dades

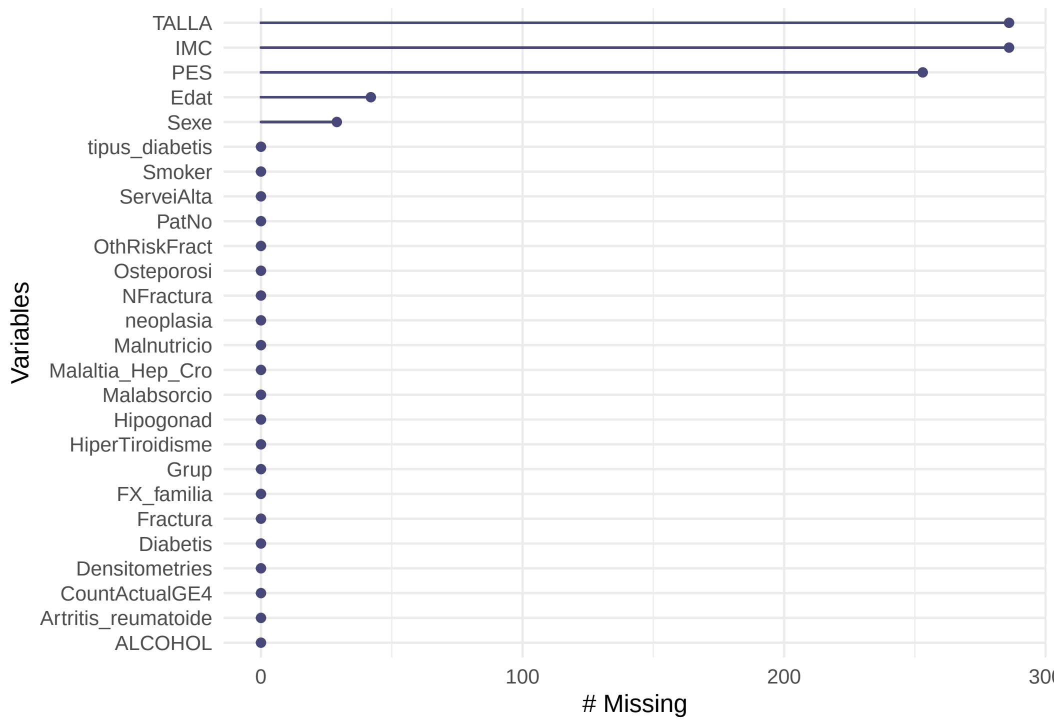
Després d'una inspecció inicial de la base de dades demogràfica[1], s'han detectat deficiències crítiques que podrien comprometre la validesa interna de l'estudi.

1.2.1 Valors absents en variables crítiques

S'ha detectat una pèrdua d'informació sistemàtica en variables que el protocol defineix com a fonamentals per a l'anàlisi

Tal com veiem a la figura[2], es detecten **registres sense informació de sexe o edat**

[1] Annex A.1
[2] Annex A.2



Per altra banda, també s'observa una mancança molt elevada de dades antropomètriques (**pes i talla**).

1.2.2 Discrepàncies de càlcul

S'ha realitzat una verificació de les variables calculades presents al fitxer i s'ha detectat que la variable `IMC` conté errors de transcripció. Al summary[3] veiem un cas de $IMC = 55$. Observant aquesta entrada d'aprop[4] veiem que el seu pes és de 55kg i talla de 152 cm.

S'han identificat altres registres on l'IMC no corresponia al càlcul manual.

1.2.3 Valors extrems i outliers

Tot i que el protocol estipula un rang d'edat de 50 a 90 anys, s'han detectat 46 pacients[5] amb edat superior als 90 anys.

[3] Annex A.3
[4] Annex A.3
[5] Annex A.4

Table 1: Comparació entre casos i controls

X	X....ALL....	X.....0.....	X.....1.....	p.overall
	N=408	N=272	N=136	
Edat	82.0 (8.84)	82.1 (8.80)	82.0 (8.97)	0.931
Sexe:				1.000
F	300 (79.2%)	200 (79.1%)	100 (79.4%)	
M	79 (20.8%)	53 (20.9%)	26 (20.6%)	
ServeiAlta:				<0.001
CGURG	31 (7.60%)	31 (11.4%)	0 (0.00%)	
GOURG	2 (0.49%)	2 (0.74%)	0 (0.00%)	
MDURG	225 (55.1%)	225 (82.7%)	0 (0.00%)	
MED	2 (0.49%)	0 (0.00%)	2 (1.47%)	
PSURG	1 (0.25%)	1 (0.37%)	0 (0.00%)	
TOURG	13 (3.19%)	13 (4.78%)	0 (0.00%)	
TRA	134 (32.8%)	0 (0.00%)	134 (98.5%)	
PES	67.4 (13.1)	68.5 (13.7)	65.0 (11.6)	0.098
TALLA	154 (8.47)	154 (8.70)	152 (7.83)	0.262
IMC	28.8 (5.62)	29.3 (5.84)	27.5 (4.90)	0.091
ALCOHOL:				0.197
0	379 (92.9%)	251 (92.3%)	128 (94.1%)	
1	26 (6.37%)	20 (7.35%)	6 (4.41%)	
2	3 (0.74%)	1 (0.37%)	2 (1.47%)	
Smoker:				0.756
0	362 (88.7%)	241 (88.6%)	121 (89.0%)	
1	15 (3.68%)	9 (3.31%)	6 (4.41%)	
2	31 (7.60%)	22 (8.09%)	9 (6.62%)	
Hipogonad: 0	408 (100%)	272 (100%)	136 (100%)	.
Artritis_reumatoide:				1.000
0	404 (99.0%)	269 (98.9%)	135 (99.3%)	
1	4 (0.98%)	3 (1.10%)	1 (0.74%)	
Fractura:				<0.001
0	296 (72.5%)	218 (80.1%)	78 (57.4%)	
1	112 (27.5%)	54 (19.9%)	58 (42.6%)	
NFractura	0.34 (0.69)	0.25 (0.64)	0.51 (0.75)	0.001
FX_familia: 0	408 (100%)	272 (100%)	136 (100%)	.
Diabetis:				0.473
0	281 (68.9%)	191 (70.2%)	90 (66.2%)	
1	127 (31.1%)	81 (29.8%)	46 (33.8%)	
tipus_diabetis:				0.433
0	281 (68.9%)	191 (70.2%)	90 (66.2%)	
1	2 (0.49%)	2 (0.74%)	0 (0.00%)	
2	125 (30.6%)	79 (29.0%)	46 (33.8%)	
Densitometries:				0.201
0	390 (95.6%)	263 (96.7%)	127 (93.4%)	
1	18 (4.41%)	9 (3.31%)	9 (6.62%)	
Osteporosi:				0.006
0	374 (91.7%)	257 (94.5%)	117 (86.0%)	
1	34 (8.33%)	15 (5.51%)	19 (14.0%)	
neoplasia:				0.311
0	358 (87.7%)	235 (86.4%)	123 (90.4%)	
1	50 (12.3%)	37 (13.6%)	13 (9.56%)	
HiperTiroidisme:				0.449

X	X....ALL....	X.....0.....	X.....1.....	p.overall
0	400 (98.0%)	268 (98.5%)	132 (97.1%)	
1	8 (1.96%)	4 (1.47%)	4 (2.94%)	
Malnutricio:				0.690
0	401 (98.3%)	268 (98.5%)	133 (97.8%)	
1	7 (1.72%)	4 (1.47%)	3 (2.21%)	
Malabsorcio:				1.000
0	407 (99.8%)	271 (99.6%)	136 (100%)	
1	1 (0.25%)	1 (0.37%)	0 (0.00%)	
Malaltia_Hep_Cro:				0.517
0	397 (97.3%)	266 (97.8%)	131 (96.3%)	
1	11 (2.70%)	6 (2.21%)	5 (3.68%)	
OthRiskFract:				0.751
0	334 (81.9%)	221 (81.2%)	113 (83.1%)	
1	74 (18.1%)	51 (18.8%)	23 (16.9%)	
CountActualGE4:				0.967
0	94 (23.0%)	62 (22.8%)	32 (23.5%)	
1	314 (77.0%)	210 (77.2%)	104 (76.5%)	

A Annex

A.1 Càrrega de dades

```
library(haven)

uab_demo1 <- read_sas("uab_demo1.sas7bdat", NULL)

uab_drugs1 <- read_sas("uab_drugs1.sas7bdat", NULL)

uab_atc_drugs1 <- read_sas("uab_atc_drugs1.sas7bdat",NULL)
library(dplyr)

vars_factor <- c("Sexe", "Grup", "ServeiAlta", "ALCOHOL", "Smoker",
                 "Hipogonad", "Artritis_reumatoide", "Fractura",
                 "FX_familia", "Diabetis", "tipus_diabetis",
                 "Densitometries", "Osteporosi", "neoplasia",
                 "HiperTiroidisme", "Malnutricio", "Malabsorcio",
                 "Malaltia_Hep_Cro","CountActualGE4","OthRiskFract")

uab_demo1 <- uab_demo1 %>%
  mutate(across(all_of(vars_factor), as.factor))
uab_demo1$Sexe[uab_demo1$Sexe == ""]=NA
```

A.2 Valors extrems

```
library(naniar)
gg_miss_var(uab_demo1)
```

A.3 Discrepàncies de càlcul

```
summary(uab_demo1)
```

```
##      PatNo        Grup          Edat           Sexe      ServeiAlta
##  Min.   :  1.0   0:272   Min.   :51.00       :  0   CGURG: 31
##  1st Qu.:102.8   1:136   1st Qu.:78.00   F   :300   GOURG:  2
##  Median :204.5           Median :84.00   M   : 79   MDURG:225
##  Mean   :204.5           Mean   :82.05   NA's: 29   MED  :  2
##  3rd Qu.:306.2           3rd Qu.:88.00              PSURG:  1
##  Max.   :408.0           Max.   :95.00              TOURG: 13
##                          NA's   :42                 TRA  :134
##       PES             TALLA             IMC         ALCOHOL  Smoker   Hipogonad
##  Min.   : 38.50   Min.   :134.0   Min.   :17.90   0:379   0:362   0:408
##  1st Qu.: 57.10   1st Qu.:147.6   1st Qu.:25.41   1: 26   1: 15
##  Median : 67.50   Median :153.0   Median :28.09   2:  3   2: 31
##  Mean   : 67.39   Mean   :153.7   Mean   :28.82
##  3rd Qu.: 74.55   3rd Qu.:160.0   3rd Qu.:31.73
```

```
##  Max.   :107.00   Max.   :179.0   Max.   :55.00
##  NA's   :253      NA's   :286     NA's   :286
##  Artritis_reumatoide Fractura   NFractura      FX_familia Diabetis
##  0:404               0:296    Min.   :0.0000   0:408      0:281
##  1:  4               1:112    1st Qu.:0.0000              1:127
##                               Median :0.0000
##                               Mean   :0.3382
##                               3rd Qu.:1.0000
##                               Max.   :7.0000
##
##  tipus_diabetis Densitometries Osteporosi neoplasia HiperTiroidisme Malnutricio
##  0:281          0:390          0:374      0:358     0:400           0:401
##  1:  2          1: 18          1: 34      1: 50     1:  8           1:  7
##  2:125
##
##
##
##
##  Malabsorcio Malaltia_Hep_Cro OthRiskFract CountActualGE4
##  0:407       0:397            0:334        0: 94
##  1:  1       1: 11            1: 74        1:314
##
##
##
##
##
```

```r
library(psych)
describe(uab_demo1)
```

```
##                      vars   n   mean     sd median trimmed    mad   min max
## PatNo                   1 408 204.50 117.92 204.50  204.50 151.23   1.0 408
## Grup*                   2 408   1.33   0.47   1.00    1.29   0.00   1.0   2
## Edat                    3 366  82.05   8.84  84.00   83.15   7.41  51.0  95
## Sexe*                   4 379   2.21   0.41   2.00    2.14   0.00   2.0   3
## ServeiAlta*             5 408   4.26   2.07   3.00    4.28   0.00   1.0   7
## PES                     6 155  67.39  13.13  67.50   66.75  12.60  38.5 107
## TALLA                   7 122 153.66   8.47 153.00  153.37   8.90 134.0 179
## IMC                     8 122  28.82   5.62  28.09   28.43   4.22  17.9  55
## ALCOHOL*                9 408   1.08   0.30   1.00    1.00   0.00   1.0   3
## Smoker*                10 408   1.19   0.55   1.00    1.02   0.00   1.0   3
## Hipogonad*             11 408   1.00   0.00   1.00    1.00   0.00   1.0   1
## Artritis_reumatoide*   12 408   1.01   0.10   1.00    1.00   0.00   1.0   2
## Fractura*              13 408   1.27   0.45   1.00    1.22   0.00   1.0   2
## NFractura              14 408   0.34   0.69   0.00    0.22   0.00   0.0   7
## FX_familia*            15 408   1.00   0.00   1.00    1.00   0.00   1.0   1
## Diabetis*              16 408   1.31   0.46   1.00    1.27   0.00   1.0   2
## tipus_diabetis*        17 408   1.62   0.92   1.00    1.52   0.00   1.0   3
## Densitometries*        18 408   1.04   0.21   1.00    1.00   0.00   1.0   2
## Osteporosi*            19 408   1.08   0.28   1.00    1.00   0.00   1.0   2
## neoplasia*             20 408   1.12   0.33   1.00    1.03   0.00   1.0   2
## HiperTiroidisme*       21 408   1.02   0.14   1.00    1.00   0.00   1.0   2
## Malnutricio*           22 408   1.02   0.13   1.00    1.00   0.00   1.0   2
## Malabsorcio*           23 408   1.00   0.05   1.00    1.00   0.00   1.0   2
```

```
## Malaltia_Hep_Cro*      24 408   1.03   0.16   1.00    1.00   0.00   1.0   2
## OthRiskFract*          25 408   1.18   0.39   1.00    1.10   0.00   1.0   2
## CountActualGE4*        26 408   1.77   0.42   2.00    1.84   0.00   1.0   2
##                        range  skew kurtosis   se
## PatNo                  407.0  0.00    -1.21 5.84
## Grup*                    1.0  0.70    -1.51 0.02
## Edat                    44.0 -1.26     1.64 0.46
## Sexe*                    1.0  1.43     0.04 0.02
## ServeiAlta*              6.0  0.35    -1.43 0.10
## PES                     68.5  0.39    -0.05 1.05
## TALLA                   45.0  0.34    -0.11 0.77
## IMC                     37.1  1.08     2.83 0.51
## ALCOHOL*                 2.0  3.95    16.23 0.01
## Smoker*                  2.0  2.75     5.93 0.03
## Hipogonad*               0.0   NaN      NaN 0.00
## Artritis_reumatoide*     1.0  9.91    96.52 0.00
## Fractura*                1.0  1.01    -0.99 0.02
## NFractura                7.0  4.21    31.04 0.03
## FX_familia*              0.0   NaN      NaN 0.00
## Diabetis*                1.0  0.81    -1.34 0.02
## tipus_diabetis*          2.0  0.82    -1.32 0.05
## Densitometries*          1.0  4.42    17.61 0.01
## Osteporosi*              1.0  3.00     7.04 0.01
## neoplasia*               1.0  2.29     3.27 0.02
## HiperTiroidisme*         1.0  6.90    45.78 0.01
## Malnutricio*             1.0  7.41    53.03 0.01
## Malabsorcio*             1.0 20.05   401.01 0.00
## Malaltia_Hep_Cro*        1.0  5.82    31.95 0.01
## OthRiskFract*            1.0  1.65     0.72 0.02
## CountActualGE4*          1.0 -1.28    -0.37 0.02
```

```r
cat("Pes =",uab_demo1[365,]$PES[1], "Talla = ", uab_demo1[365,]$TALLA[1], "IMC = ", uab_demo1[365,]$IMC
```

```
## Pes = 55 Talla =  152 IMC =  55
```

```r
real= uab_demo1[365,]$PES[1]/(uab_demo1[365,]$TALLA[1]/100)^2; real
```

```
## [1] 23.8054
```

A.4 Valors extrems

```r
length(which(uab_demo1$Edat>90))
```

```
## [1] 46
```